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ABSTRACT

We have obtained a Late Permian paleomagnetic pole from dikes in the Tarim craton of China. Directions of magnetization from 21 mafic and ultramafic dikes at three localities pass a fold test at the 95% confidence level. The new paleomagnetic pole lies at lat 66°N, long 181° E; $N = 21$, $K = 61$, and $\alpha_{95} = 3.9°$. On the basis of a comparison with paleomagnetic data from adjacent tectonic units, we conclude that (1) the Late Permian paleolatitude of the Tarim craton differs from that of the Sino-Korean craton, suggesting that the two cratons were distinct tectonic units in Late Permian time; (2) there was relative motion between the Tarim craton and the Siberian craton after Late Permian time; and (3) negligible relative motions have occurred between different thrust sheets in the Keping-Bachu area of the northwestern Tarim craton.

*Deceased.

INTRODUCTION

In its simplest form, China consists of the Sino-Korean, Yangtze, and Tarim cratons, which are separated by Paleozoic and Mesozoic fold belts (Fig. 1). Western China consists mostly of the Tarim craton in the north and numerous accreted terranes to the south. The Tarim craton is between the Tianshan Range to the north and west and the Kunlun-Altyn Mountains to the south. The central part of the craton is thought to be an Archean nucleus, now covered by Proterozoic and Phanerozoic strata. The strata exposed on the periphery of the craton are clastics, volcanics, and tillites of the upper Precambrian Sinian System, overlain by Paleozoic carbonates and marine clastics, Mesozoic-Cenozoic redbeds, and localized limestone in the western part of the Tarim Basin.

Two general models exist for the formation of the Tarim craton. (1) On the basis of the observation that the main part of the basement of the Tarim craton was consolidated prior to the Middle Proterozoic (Li et al., 1980; Wang and Liu, 1981), the Tarim craton formed at about the same time as the Sino-Korean craton. Thus, the Sino-Korean and Tarim

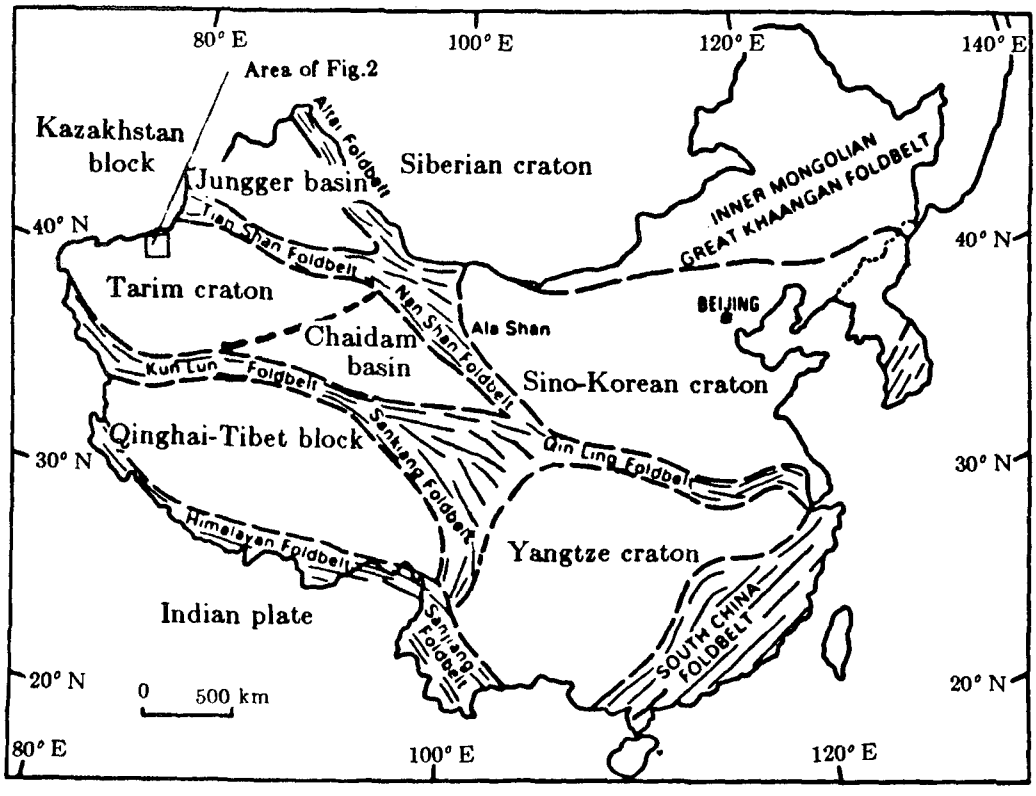


Figure 1. Major tectonic units of Asia.

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cratons are often treated as a single tectonic unit called the Tarim–Sino-Korean craton (Li et al., 1980; Zhang et al., 1984). (2) On the basis of lithological and faunal similarities (Huang, 1978, Gao and Wu, 1984), the Tarim craton was closely related to the Yangtze craton in the early Paleozoic.

Field evidence suggests that the Tarim craton was joined with the Tianshan terrane in the Permian (Wang and Liu, 1981; Zhang et al., 1984; Chen and Liang, 1985; Zhang and Wu, 1985). Because the Tarim craton is the innermost of the three major cratons, it must have arrived at its present position relative to the Siberian craton at least as early as any of the outer cratons and associated terranes. We collected paleomagnetic samples from the Tarim craton to help illustrate its tectonic relation with adjacent tectonic blocks.

In the summer of 1986, a paleomagnetic study of the Tarim craton was undertaken jointly by members of the Xinjiang Geology Research Institute, Beijing Graduate School of Wuhan Geology College, Chinese Academy of Geological Sciences, and the Geophysics Department at Stanford University. Here we report new Late Permian paleomagnetic data as the first of our results.

GEOLOGIC SETTING

The Keping-Bachu area is on the northwestern rim of the Tarim craton (Figs. 1 and 2). The Paleozoic sequences in this area are virtually continuous, with the exception of a few disconformities. In a simplified view, there are two phases of tectonic activity. The first is Permian in age; it is characterized by eruption of dikes and basalts in the Keping-Bachu area on the northern margin of the craton and by uplift from marine or shallow-marine environment to continental environment. The second phase occurred during the Cenozoic and is probably related to the collision of the Indian plate with the Eurasian plate. The latter phase is characterized by tilting and folding around the margins of the craton. The most obvious tectonic features in the sampling area are multiple thrust sheets, which detach Sinian to Mesozoic stratigraphic packages from the basement. These thrust structures could, in principle, produce relative rotations between different thrust blocks; therefore, our strategy was to sample as many thrust sheets as possible so that any local rotations could be detected.

We believe that within each locality the dikes sampled are coeval. A K/Ar age of 303 Ma (Xinjiang Second Geological Brigade, 1985, unpub. rept.) on hornblende-syenite samples from Bachu is clearly incorrect because the dikes cut Lower Permian limestone at one locality in Bachu. However, the dikes do not cut Lower Jurassic strata on a regional basis. Therefore, it is widely accepted in Xinjiang that the dikes are Late Permian

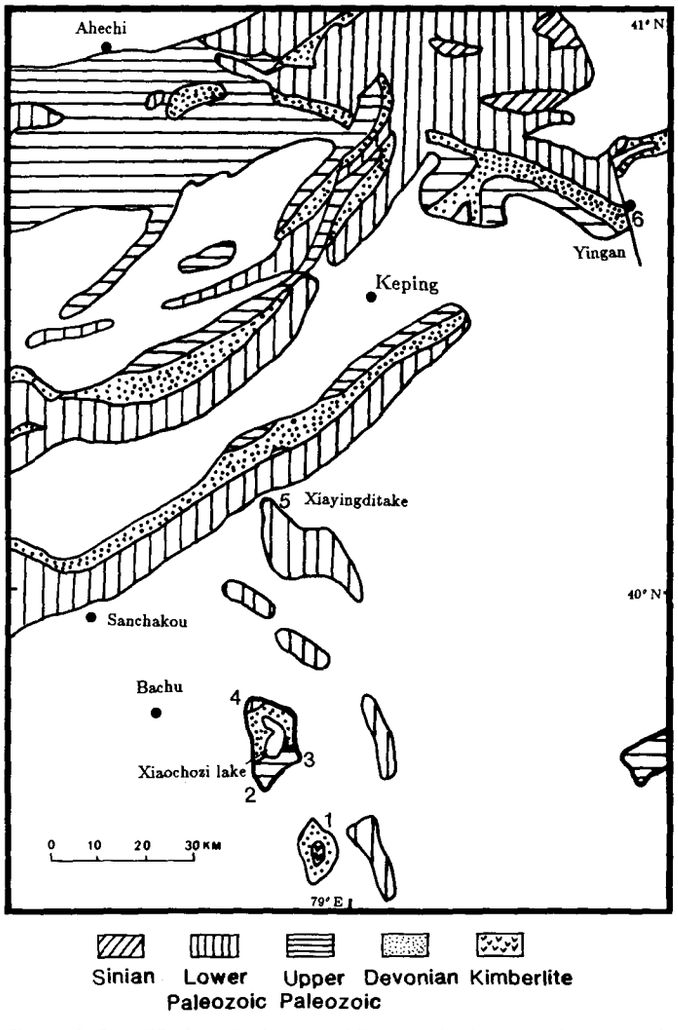


Figure 2. Simplified geologic map of Keping-Bachu area of northwestern Tarim craton, Xinjiang, China. Numbers indicate paleomagnetic sampling localities.

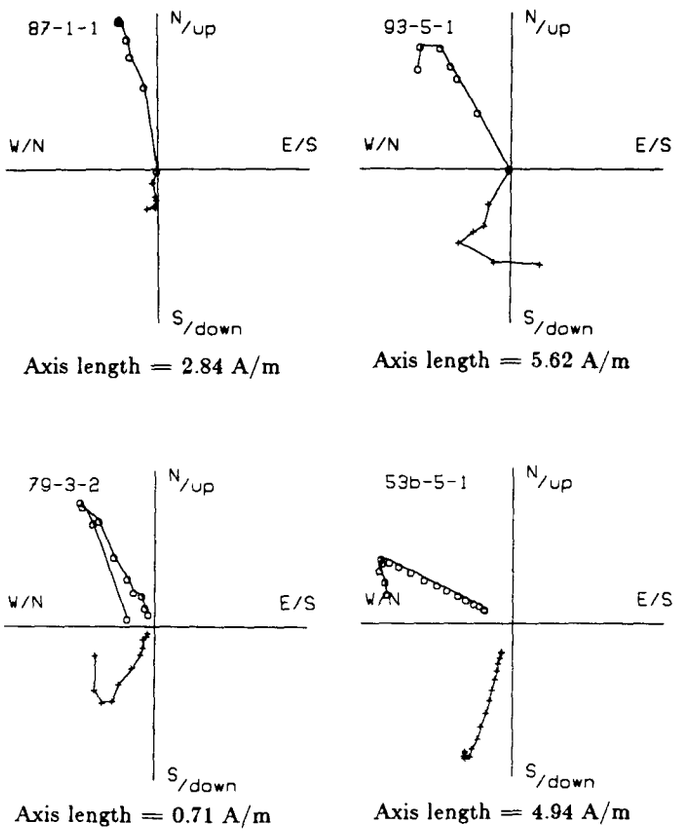


Figure 3. Orthogonal vector plots of four representative samples from Upper Permian dikes. Demagnetization steps plotted for samples 87-1-1 and 93-5-1 are NRM, 139, 350, 500, 550, 575, and 600 °C; for sample 79-3-2, steps plotted are NRM, 5, 10, 15, 20, 30, 40, 50, 60, 70, and 80 mT; and for sample 53b-5-1, steps plotted are NRM, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, and 80 mT. Zero is projection on horizontal plane; plus symbol is projection on vertical, north-south plane. All plots are in geographic coordinates.

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in age because the Tarim craton was tectonically active in the late Paleozoic.

PALEOMAGNETIC STUDY

We collected samples at three localities: Bachu (lat 39.8°N, long 78.8°E; 1, 2, 3, and 4 in Fig. 2), Yinggan (lat 40.6°N, long 79.5°E; 6 in Fig. 2), and Xiayingditake (lat 40.2°N, long 78.8°E; 5 in Fig. 2). Sun-compass orientations were used for most samples. The dikes cut the Devonian redbeds at Yinggan and Bachu and the Ordovician limestones at Xiayingditake. Corrections for tectonic tilt were made according to the attitude of the country rocks because dikes cut the country rocks vertically. All sites except two at Yinggan show only small changes in directions after tilt corrections because dips were 15° or less. However, the dikes at Yinggan intruded the flat-lying Upper Devonian redbeds before being tilted 45° to the south; knowing this allowed us to perform a fold test.

All sites consist of six cores, each core yielding at least one specimen. Specimens were treated by alternating field (AF) and thermal demagnetization. AF demagnetization was as efficient as thermal demagnetization in isolating stable magnetizations, and therefore most specimens were demagnetized by AF means. After removal of a viscous remanent magnetization parallel to the present Earth's field direction in low fields or temperatures, most specimens show single-component behavior (Fig. 3). All specimens contain a reversed magnetization consistent with the Kiaman reversed epoch (248–315 Ma). Site mean magnetization directions are converted to VGP (virtual geomagnetic pole) positions in Table 1 so that results from different localities may be compared. It is apparent that the VGP values are better grouped after tilt correction than before; this suggests a positive fold test (K_a/K_b = 1.96: K is a precision parameter; Fisher, 1953) at the 95% confidence level, according to the test of McElhinny (1964) (Fig. 4). The arguments presented by McFadden and Jones (1981) indicate that the fold test by McElhinny's method makes a conservative estimate of probability. This positive fold test confirms that the magnetization isolated in this study is of prefolding or pre-Tertiary origin.

DISCUSSION

There is only one published paleomagnetic study of the Tarin craton apart from our work. Bai et al. (1987) produced paleomagnetic poles for the Upper Devonian, upper Carboniferous, and Lower Permian rocks from an area about 200 km east of our main localities. Their Early Permian pole is based on data from 26 sites of basalt and limestone. Thermal demagnetization experiments were successful in removing younger overprints from the Lower Permian samples.

Our new pole from the Upper Permian dikes does not coincide with the Early Permian pole reported by Bai et al. (1987), but it is on a moving

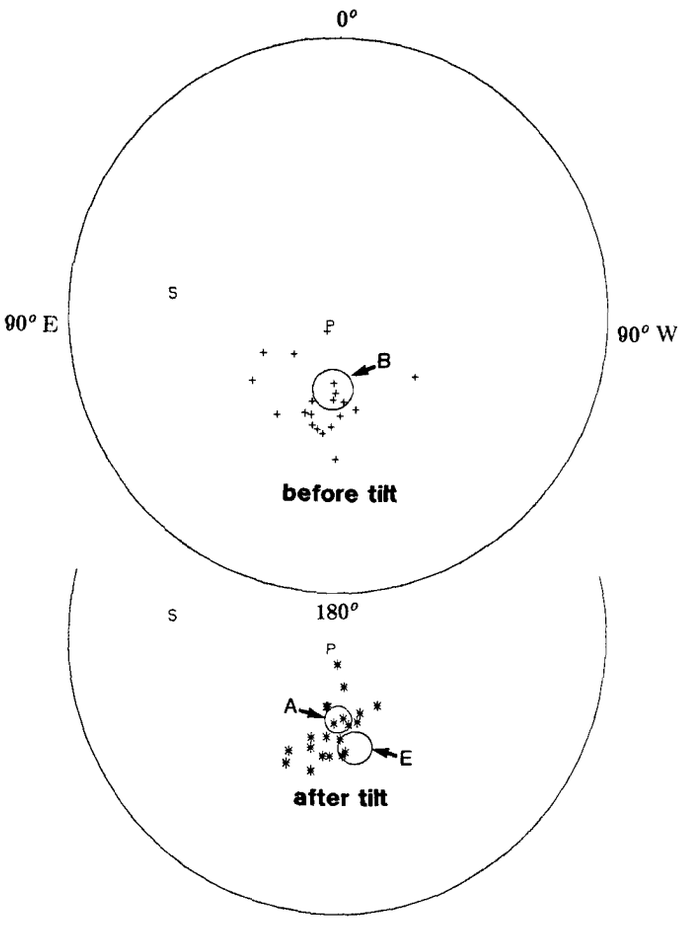


Figure 4. Equal-area stereonet diagrams of VGP (virtual geomagnetic pole) from each site mean direction of Upper Permian dikes. Plus symbols indicate before tilt correction; star symbols, after tilt correction. B and A: mean of VGP before and after tilt correction. E: Early Permian pole of Bai et al. (1987). S: sampling site in Tarim craton. P: present north pole.

TABLE 1. PALEOMAGNETIC DATA OF UPPER PERMIAN DIKES OF TARIM CRATON, XINJIANG

Site*	N+	Pole position (in situ) lat° N	(in situ) long° E	(corrected) lat° N	(corrected) long° E	K	α_{95}
52	3(s)	56.8	241.0	59.3	188.9	236	5.3
53	3(s)	59.5	239.5	63.7	178.2	214	5.5
79	6	63.4	163.8	58.3	179.7	408	2.8
87	7	67.8	112.2	52.8	158.7	100	5.3
88	8	87.8	152.0	69.2	206.4	85	5.4
89	6	60.8	125.0	56.8	157.6	354	3.0
90	6	76.1	128.0	72.8	178.2	69	6.9
91	8	59.1	149.2	53.4	170.9	364	2.6
92	7	64.2	197.8	68.0	220.8	77	6.0
93	6	67.3	166.8	78.4	201.4	143	4.8
94	6	58.6	172.7	69.0	190.9	479	2.6
95	6	62.7	168.0	62.7	168.0	187	4.2
96	6	62.9	187.4	62.9	187.4	167	4.4
97	7	67.5	183.4	67.5	183.4	507	2.5
98	6	50.4	182.6	58.0	182.6	154	4.6
99	6	60.0	181.2	67.2	194.3	296	3.3
100	6	57.8	175.9	57.8	175.9	1083	1.7
101	6	70.2	184.6	85.1	204.0	139	4.9
102	6	60.1	169.1	60.1	169.1	199	4.1
103	6	67.2	190.5	67.2	190.5	109	5.5
104	6	73.2	184.4	73.2	184.4	77	6.5
Mean	21(s)	68.0	177.0	-	-	31	5.5
Mean	21(s)	-	-	65.6	181.2	61	3.9

* Sites 52 and 53 are located at lat 40.6° N, long 79.5° E; three sites (6 cores /site) are from each dike at this locality. Site 79 is located at lat 40.2° N, long 78.8° E. Sites 87 to 104 are located at lat 39.8° N, long 78.8° E.

+ Number of core or site (s).

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trend defined by paleomagnetic poles from the Devonian to the Early Permian (Fig. 5; Bai et al., 1987). According to both geologic and paleomagnetic evidence, the Tarim craton had collided with the Kazakhstanian/Tianshan block by late Carboniferous time (Bai et al., 1987). Compared with the Early Permian pole of Bai et al. from the Tarim craton, this suggests that the northward translation of the Tarim craton had not ended after the initial collision. Analogies with the collision of India and Tibet seem particularly appropriate. The initial collision between India and Asia was followed by significant crustal shortening for at least 30 to 40 m.y. (Besse et al., 1984).

Our sampling sites are distributed on four separate thrust sheets; the entire sampling area spans 60 000 km^2. The paleomagnetic directions from all sites show no significant differences, yet all have reversed polarity, suggesting (1) small, if any, relative motions between different thrust sheets, and (2) coeval dikes (Kiaman age).

By choosing a reference site at the center of the Tarim craton (lat 37.5°N, long 82.5°E), the expected Late Permian paleolatitude is 30.4° on the basis of our new paleomagnetic pole. However, the Late Permian paleolatitude expected for a reference site (lat 37.5°N, long 115°E) at the center of the Sino-Korean craton is 15.4° on the basis of an average of data for six Late Permian poles (Zhao and Coe, 1987). The paleolatitude of the Tarim craton is quite different from the paleolatitude of the Sino-Korean craton, suggesting that these two cratons were not in the same latitudinal area in the Late Permian.

The Late Permian pole of the Tarim craton is also different from that of the Siberian craton (Fig. 5), suggesting either that the Tarim craton had not terminally collided with the Siberian craton by the Late Permian or that the Tarim craton moved after collision. Because all known late Paleozoic poles from the Tarim craton are to the east of those from the Siberian craton (Bai et al., 1987), it is likely that the Tarim craton moved relative to the Siberian craton after Permian time. It has been suggested that several left-lateral strike-slip faults in central Asia are a consequence of the collision between India and the Eurasian plates (Tapponnier et al., 1982; Lin et al., 1985). As a result, the Yangtze craton (south China block) and the Chaidam terrane may have moved eastward along these faults (Lin et al., 1985; Li et al., 1986). On the basis of the paleomagnetic data available from the Tarim craton and the Siberian craton, we believe that the Tarim craton moved east relative to the Siberian craton. The timing and displacement of motion remain unknown until other paleomagnetic data are available from the Tarim craton.

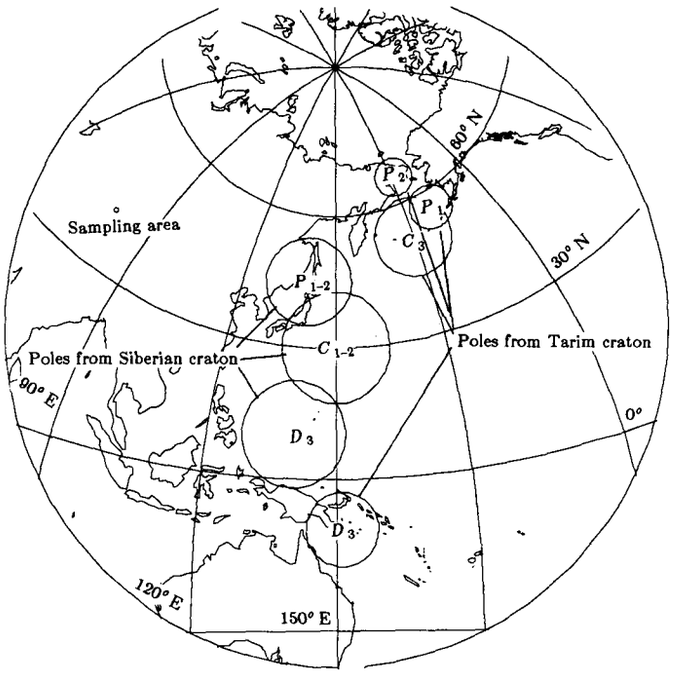


Figure 5. Polar equal-area map of paleomagnetic poles from Tarim craton and Siberian craton (Khramov et al., 1985). Poles from Siberian craton: D_3, Late Devonian; C_{1-2}, early to middle Carboniferous; P_{1-2}, Early to middle Permian. Poles from Tarim craton: D_3, Late Devonian; C_3, late Carboniferous; P_1, Early Permian; P_2, Late Permian. Poles of D_3, C_3, and P_1 for Tarim craton are from Bai et al. (1987); P_2 is from this study.

REFERENCES CITED

Bai, Y., Chen, G., Sun, Q., Sun, Y., Li, Y., Dong, Y., and Sun, D., 1987, Late Paleozoic polar wander path for the Tarim platform and its tectonic significance: Tectonophysics, v. 139, p. 145–153.

Besse, J., Courtillot, J., Pozzi, P., Westphal, M., and Zhou, Y., 1984, Paleomagnetic estimates of crustal shortening in the Himalayan thrusts and Zangbo suture: Nature, v. 311, p. 621–626.

Chen, Z., and Liang, Y., 1985, Research on several problems of structural geology in Tianshan, Xinjiang: Xingjiang Geology, v. 3, p. 1–13 (in Chinese with English abstract).

Fisher, R.A., 1953, Dispersion on a sphere: Royal Society of London Proceedings, v. A217, p. 295–305.

Gao, Z., and Wu, S., 1984, Tectonic development of Precambrian-Palaeozoic Tarim Massif: Kexue Tongbao, v. 29, p. 1523–1526 (in Chinese with English abstract).

Huang, T.K., 1978, An outline of the tectonic characteristics of China: Eclogae Geologicae Helvetiae, v. 71, p. 611–635.

Khramov, A.N., Gurevich, E.L., Komissarova, R.A., Osipova, E.P., Pisarevsky, S.A., Rodionov, V.P., and Slautsitais, I.P., 1985, Paleomagnetism, microplates and Siberian plate consolidation: Journal of Geodynamics, v. 2, p. 127–139.

Li, C.Y., Wang, Q., Zhang, Z., and Liu, X., 1980, A preliminary study of plate tectonics of China: Chinese Academy of Geological Science Bulletin, v. 2, p. 11–22 (in Chinese with English abstract).

Li, Y., McWilliams, M., and Tan, C., 1986, New paleomagnetic results from the terranes of western China: EOS (American Geophysical Union Transactions), v. 67, p. 925.

Lin, J., Fuller, M., and Zhang, W., 1985, Preliminary Phanerozoic polar wander paths for the north and south China blocks: Nature, v. 313, p. 444–449.

McElhinny, M.W., 1964, Statistical significance of the fold test in paleomagnetism: Royal Astronomical Society Geophysical Journal, v. 8, p. 338–340.

McFadden, P.L., and Jones, D.L., 1981, The fold test in paleomagnetism: Royal Astronomical Society Geophysical Journal, v. 67, p. 53–58.

Tapponnier, P., Peltzer, A., Dain, A., Armijo, R., and Cobbold, P., 1982, Propagating extrusion tectonics in Asia: New insights from simple experiments with plasticine: Geology, v. 10, p. 611–616.

Wang, H., and Liu, B., 1981, Paleogeographic frame of China since the Middle Proterozoic: Journal of Stratigraphy, v. 5, p. 77–89 (in Chinese with English abstract).

Zhang, L., and Wu, N., 1985, The geotectonics and the evolution of Tianshan: Xingjiang Geology, v. 3, p. 1–14 (in Chinese with English abstract).

Zhang, Z.M., Liou, J.G., and Coleman, R.G., 1984, An outline of the plate tectonics of China: Geological Society of America Bulletin, v. 95, p. 295–312.

Zhao, X., and Coe, R.S., 1987, Paleomagnetic constraints on the collision and rotation of North and South China: Nature, v. 327, p. 141–144.

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